

An Improved Visual Enhancement Method for Color Images

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ABSTRACT

The paper presents an improved method for color image enhancement based on dynamic range compression and shadow compensation method proposed by Felix Albu et al. The improved method consists of not only the dynamic range compression and shadow compensation but also intensity contrast stretching implemented under the logarithmic image processing (LIP) model. The previous method enhance image while preserve image details and color information without generating visual artifacts. On the premise of remaining the advantages of the previous one, our improved method enhances the intensity of the whole image especially the low-light areas. The experimental results illustrate the effectiveness of our proposed method and the superiority over the previous one.

Keywords: Color image enhancement, Logarithmic image processing, Dynamic range compression, Shadow compensation, Intensity contrast stretching

1. INTRODUCTION

Color image, which has been widely used in many areas such as medical imaging and aerial photograph, contain much more information than gray-scale images. Therefore, the enhancement of color image, which aims to improve the image quality for human visual perception, is of great significant in image analysis and process. The most popular method for gray-scale image enhancement is histogram equalization [1], which is based on the assumption that a uniformly distributed gray-scale histogram will have the best visual contrast. With the concept of histogram equalization, Trahanias and Venetsanopoulos [2] proposed a uniform 3-D histogram specification method in the RGB color space. Felix Albu and Constantin Vertan [3] proposed a technique of dynamic range compression and shadow compensation for color images. It works well on enhancing image preserving the colors and details. To achieve the algorithms, color image enhancement is formulated as an optimization problem.

The existing methods mentioned above have several drawbacks. The uniform 3-D histogram specification method in the RGB color space didn't take the physical image formation model and the human visual system into consideration. The dynamic range compression and shadow compensation method was not effective enough in dark areas. The logarithmic image processing model proposed by Jourlin and Pinoli [4-7] provides a mathematical framework which is compatible with the physical image formation model and human visual system. Therefore, we extend the LIP model in RGB color space and improve the dynamic range compression and shadow compensation method combining intensity contrast stretching method applied in the RGB color space. The results obtained from the improved method are compared with that obtained from the traditional one and the uniform 3-D histogram specification method.

The reminder of the paper is organized as follow: Related works are described in Section 2. Our proposed method based on LIP model is proposed in Section 3. The experimental results are illustrated in Section 4 and the conclusion is given in Section 5.

2. RELATED WORKS

2.1. Logarithmic Image Processing Model

The logarithmic image processing (LIP) framework, which is based on physical relevant image formation model, open a new way to the development of image processing technique by providing specific operations and structures for

image presentation and processing. In the LIP model [7], an intensity image F is represented by its associated gray tone function f . Such a function is defined on a non-empty spatial domain D in the Euclidean space R^2 , called the spatial support, with values in the bounded real number interval $[0, M)$, where M is strictly positive (in the digital case, M equal 256 for an 8-bit image). The value of a gray tone function at a spatial location (i, j) is called a gray tone, and the real number range interval $[0, M)$ is thus called the gray tone range.

2.2. Dynamic Range Compression and Shadow Compensation Method

According to Felix Albu and Constantin Vertan [3], the dynamic range compression and shadow compensation method enhance the images with adaptive, pixel-specific color component amplification factors that are computed according to the local contrast and pre-computed setup parameters. The original global contrast is increased in the case of low-contrast, shadowed images and kept at near constant values in the remainder of the cases. The amplification is performed with a non-linear, perceptual-resembling logarithmic multiplication. And by using a recursive filter, the method can be implemented as a one-pass image processing operation. In addition, it is very suitable for implementation on low power, low memory embedded devices for using local image information and processing the image in a single scan.

3. PROPOSED METHOD

The dynamic range compression and shadow compensation method works well on enhancing low-light areas while preserving the color information and details of image, without generating visual halos. But it has drawback for the dark areas are not enhanced enough. Thus, we improve the traditional method by combining it with intensity contrast stretching method applied in the LIP model. Thus, our proposed method consists of two main steps: the dynamic range compression and shadow compensation and the intensity contrast stretching.

We firstly perform the dynamic range compression and shadow compensation method.

For each pixel, located at coordinate (i, j) , $\bar{R}(i, j), \bar{G}(i, j), \bar{B}(i, j)$ are computed by the following equation:

$$\begin{aligned}\bar{R}(i, j) &= \beta \bar{R}(i, j-1) + (1-\beta)I(i, j,1) \\ \bar{G}(i, j) &= \beta \bar{G}(i, j-1) + (1-\beta)I(i, j,2) \\ \bar{B}(i, j) &= \beta \bar{B}(i, j-1) + (1-\beta)I(i, j,3)\end{aligned}\tag{1}$$

Where $I(i, j,1), I(i, j,2), I(i, j,3)$ are color components of color image $I(i, j)$, and β is a tuning coefficient with possible values in the range $[0, 1]$. Then two correction terms for each pixel, γ_R and γ_B , are computed by the following equation for considering G component as reference.

$$\begin{aligned}\gamma_R(i, j) &= \frac{\bar{G}(i, j)}{\bar{R}(i, j)} \left(\frac{(1-\alpha)\bar{R}(i, j) + 255\alpha}{(1-\alpha)\bar{G}(i, j) + 255\alpha} \right) \\ \gamma_B(i, j) &= \frac{\bar{G}(i, j)}{\bar{B}(i, j)} \left(\frac{(1-\alpha)\bar{B}(i, j) + 255\alpha}{(1-\alpha)\bar{G}(i, j) + 255\alpha} \right)\end{aligned}\tag{2}$$

Where γ_R and γ_B are limited to $[0.95 \dots 1.05]$, and α is a tuning coefficient with possible values in the range $[0, 1]$. The modified pixel values $F(i, j)$ are computed as follow:

$$\begin{aligned}F(i, j,1) &= \gamma_R I(i, j,1) \\ F(i, j,2) &= I(i, j,2) \\ F(i, j,3) &= \gamma_B I(i, j,3)\end{aligned}\tag{3}$$

The shadow correction aims to compute amplification factors for next step pixel amplification. For each pixel, the amplification factor $H(i, j)$ is computed as follow:

$$H(i, j) = p * H(i, j-1) + (1-p) \left((1-\alpha) + \frac{255\alpha}{Y(i, j)} \right)\tag{4}$$

Where p is the pole of the recursive filter and is limited to $[0,1]$. And $Y(i, j)$ is luminance of corresponding pixel located at coordinate (i, j) .

The final modified image $F_{out}(i, j)$ is given by equation in LIP model.

$$F_{out}(i, j, k) = M - M \left(1 - \frac{F(i, j, k)}{M} \right)^{H(i, j)}, k = 1, 2, 3 \quad (5)$$

The three color components all perform pixel amplification with amplification factors $H(i, j)$. With the logarithmic image processing model the pixel amplification is ensured range preservation of the image components because of the closing property.

Secondly, we perform the intensity contrast stretching.

We define the intensity component in the RGB color space as Strickland, Kim and McDonnell [8] suggested using the intensity component that can be compatible with the human visual perception. Therefore, the intensity component $L(i, j)$ is defined as follow:

$$L(i, j) = 0.299 \times F_r(i, j) + 0.587 \times F_g(i, j) + 0.114 \times F_b(i, j) \quad (6)$$

The gray tone function of the intensity component is given by $l(i, j) = M - L(i, j)$. And, there is an optimal overall logarithmic gain which can maximally enlarge the dynamic range of the intensity component [9]. The optimal overall logarithmic gain λ_i is computed as follow:

$$\lambda_i = \frac{\ln\left(\frac{(M - l_{\max}) / M}{(M - l_{\min}) / M}\right)}{\ln((M - l_{\min}) / (M - l_{\max}))} \quad (7)$$

Where $l_{\max}$ and $l_{\min}$ designate the lower bound and the upper bound of l respectively. The intensity contrast stretching operator λ in RGB color space is given by:

$$\lambda = [\lambda_r, \lambda_g, \lambda_b]^T \quad (8)$$

Finally, the enhanced color image $F_{final}(i, j)$ is computed as follow:

$$F_{final}(i, j, k) = M - M \left(1 - \frac{f_{out}(i, j, k)}{M} \right)^{\lambda_i}, k = 1, 2, 3 \quad (9)$$

4. EXPERIMENTAL RESULTS

In this Section, the proposed method is compared with the uniform 3-D histogram specification approach and the dynamic range compression and shadow compensation method. Fig.1 and Fig.2 show the performance comparison of our improved dynamic range compression and shadow compensation method.

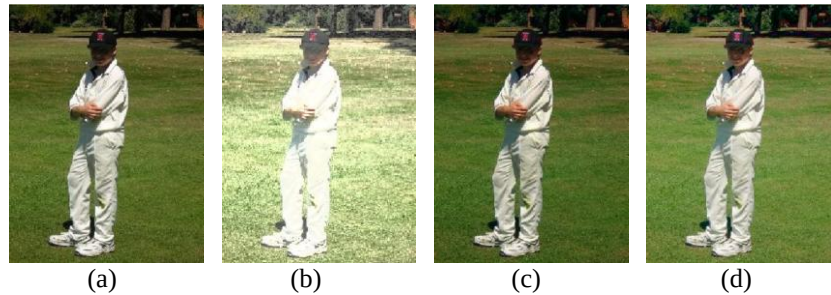


Fig 1. The performance comparison of proposed method. (a): the original image of “cricketer”; (b): the 3-D histogram specification method; (c): the enhanced image by the dynamic range compression and shadow compensation method; (d): the enhanced image by our proposed method

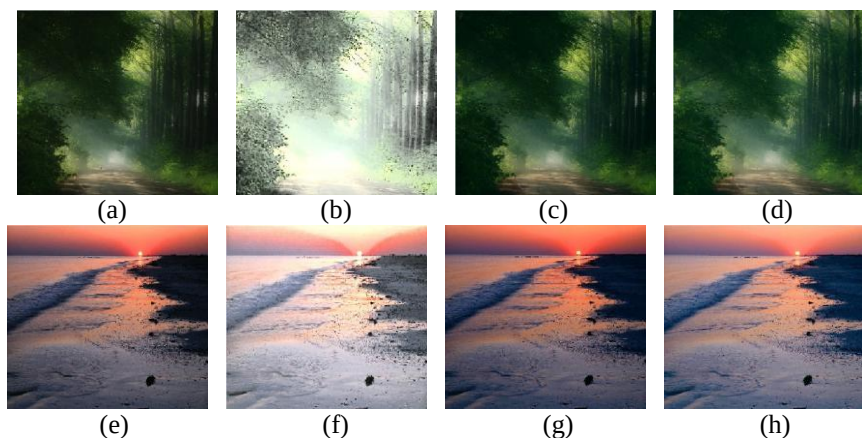


Fig 2. The performance comparison of proposed method. (a)(e): the original images of “forest” and “beach”; (b)(f): the enhanced images by 3-D histogram specification method; (c)(g): the enhanced images by the dynamic range compression and shadow compensation method; (d)(h): the enhanced images by our proposed method

Three test images are applied to evaluate the performance of proposed method and other methods that are presented here. From the experimental results, we can obtain that the results of our method look much better and more natural than that by any of the two existing methods for comparison. It can also be found that the images become lighter and the detail information become more noticeable. In addition, the low-light areas are enhanced and the colors information is preserved. The experimental results fully demonstrate the effectiveness of our proposed method.

5. CONCLUSION

This paper presents an improved method for color image enhancement. The proposed method improved the dynamic range compression and shadow compensation method by combining the intensity contrast stretching method which is applied under the LIP model. According to the simulation results obtained from the experiments, our improved enhancement method can enhance all areas of image especially the low-light areas, while at the same time preserving image details and color information without generating visual artifacts, which is suitable for human visual perception.

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